



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 3 Multiple Choice

9189/3

JUNE 2018 SESSION

1 hour

Additional materials:

- Data Booklet
- Mathematical tables and/or calculator
- Multiple Choice answer sheet
- Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 13 printed pages and 3 blank pages.

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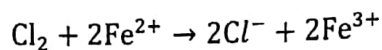
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Section A

For each question, there are four possible answers, A, B, C and D. Choose the one you consider to be correct.

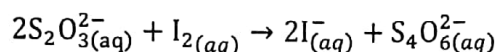
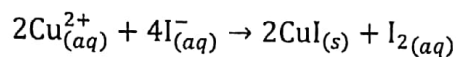
- 1 Chlorine gas and iron (II) ions react in aqueous solution as shown:



What is the E_{cell}^θ for the reaction?

- A +0.59 V
- B +1.80 V
- C +1.40 V
- D +0.92 V

- 2 A mass of 1.249 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ was dissolved and added to an excess of $\text{KI}_{(\text{aq})}$. The iodine liberated was titrated against aqueous sodium thiosulphate.



What volume of $0.100 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_{3(\text{aq})}$ was needed to react with the iodine liberated?

- A 10 cm^3
- B 25 cm^3
- C 50 cm^3
- D 100 cm^3

- 3 A sample of oxygen gas containing isotopes of mass numbers 16 and 18 was analysed in a mass spectrometer.

How many peaks corresponding to O_2^+ were recorded?

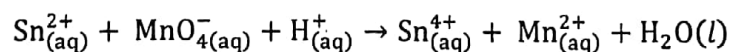
- A 1
- B 2
- C 3
- D 4

- 4 Which one is **not** a property of a real gas?
- A There are intermolecular forces between the gas particles.
 - B The volume of the gas particles is negligible.
 - C The volume of the gas particles is not negligible.
 - D The pressure exerted by the gas particles is negligible.
- 5 A volume of 100 cm^3 of hydrocarbon X was burnt in 500 cm^3 of oxygen. 50 cm^3 of oxygen remained unused. 300 cm^3 of carbon dioxide and 300 cm^3 of water vapour were formed.

What is the molecular formula of X?

- A C_2H_6
 - B C_3H_6
 - C C_3H_8
 - D C_4H_{10}
- 6 What is the correct shape of the NO_2^+ ion?
- A non-linear
 - B pyramidal
 - C planar
 - D linear
- 7 What volume of oxygen remains when 40 cm^3 of ammonia burns in 70 cm^3 of oxygen to produce nitrogen and water at s.t.p?
- A 25 cm^3
 - B 35 cm^3
 - C 40 cm^3
 - D 45 cm^3
- 8 Which particle will have a half filled P-orbital when it loses an electron?
- A P
 - B P^-
 - C Si
 - D S^{2+}

- 9 The unbalanced equation for the reaction of $\text{Sn}_{(\text{aq})}^{2+}$ and $\text{MnO}_{4(\text{aq})}^-$ is:



What is the ratio of $\text{Sn}^{2+} : \text{MnO}_4^-$ if the equation is correctly balanced?

- A 1:1
- B 2:5
- C 1:2
- D 5:2

- 10 A volume of 200 cm^3 of chlorine gas at r.t.p was passed into 100 cm^3 of $0.200 \text{ mol dm}^{-3}$ potassium iodide to produce iodine as shown in the equation.

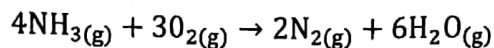


How much iodine was produced?

- A 1.06 g
 B 2.12 g
 C 4.24 g
 D 5.08 g
- 11 The equation $\text{K}_{(\text{s})} \rightarrow \text{K}^{+}_{(\text{g})} + \text{e}^{-}$ represents the
- A enthalpy change of vaporisation of potassium.
 B first ionisation energy of potassium.
 C sum of enthalpy change of atomisation and first ionisation energy of potassium.
 D sum of first ionisation energy and electron affinity of potassium.
- 12 Which equation correctly defines enthalpy change of formation of carbon monoxide?
- A $\text{C}_{(\text{s})} + \frac{1}{2}\text{O}_{2(\text{g})} \rightarrow \text{CO}_{(\text{g})}$
 B $\text{C}_{(\text{s})} + \text{O}_{(\text{g})} \rightarrow \text{CO}_{(\text{g})}$
 C $\text{C}_{(\text{s})} + \text{CO}_{2(\text{g})} \rightarrow 2\text{CO}_{(\text{g})}$
 D $\text{C}_{(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})} \rightarrow \text{CO}_{(\text{g})}$
- 13 Which equation defines the lattice energy of the ionic compound, XY?
- A $\text{X}_{(\text{s})} + \text{Y}_{(\text{s})} \rightarrow \text{XY}_{(\text{s})}$
 B $\text{X}_{(\text{g})} + \text{Y}_{(\text{g})} \rightarrow \text{XY}_{(\text{s})}$
 C $\text{X}^{+}_{(\text{s})} + \text{Y}^{-}_{(\text{s})} \rightarrow \text{XY}_{(\text{s})}$
 D $\text{X}^{+}_{(\text{g})} + \text{Y}^{-}_{(\text{g})} \rightarrow \text{XY}_{(\text{s})}$

- 14 A current of 4 A was passed through molten aluminium oxide for 200 minutes.
What is the volume of the gas liberated at s.t.p.?
- A 2.8 dm^3
B 5.6 dm^3
C 8.4 dm^3
D 11.2 dm^3
- 15 The solubility of magnesium hydroxide is $x \text{ mol dm}^{-3}$.
What is the solubility product of magnesium hydroxide?
- A $3x^2$
B $4x^3$
C $2x^2$
D $4x^2$
- 16 In the reaction, $X + Y \rightarrow 2Z$, it was found that doubling the concentration of X caused the rate of reaction to increase four times but doubling the concentration of Y did not affect the rate.
What is the rate expression for this reaction?
- A $k [X]^4$
B $4 k [X]^2$
C $4k[X]$
D $k [X]^2$
- 17 What is the maximum mass of copper formed at the cathode when 3 moles of electrons are passed through molten copper salt?
- A 42.33 g
B 95.25 g
C 127.00 g
D 190.50 g

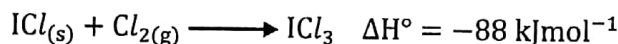
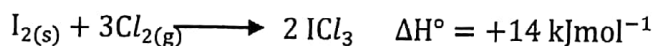
- 18 The reaction shows the oxidation of ammonia.



The rate of formation of $\text{N}_{2(g)}$ was found to be $0.27 \text{ mol dm}^{-3} \text{ s}^{-1}$.

At what rate was water being formed?

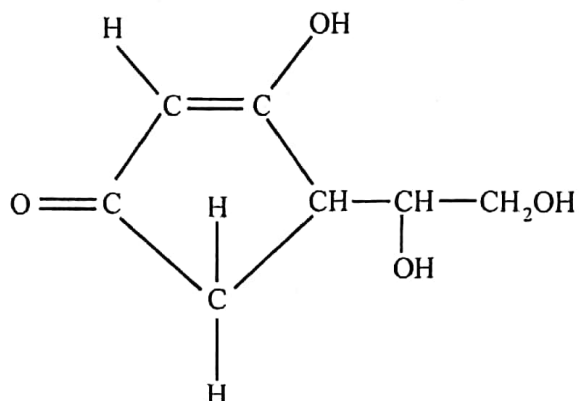
- A $0.27 \text{ mol dm}^{-3} \text{ s}^{-1}$
 - B $0.135 \text{ mol dm}^{-3} \text{ s}^{-1}$
 - C $0.54 \text{ mol dm}^{-3} \text{ s}^{-1}$
 - D $0.81 \text{ mol dm}^{-3} \text{ s}^{-1}$
- 19 Which compound has the largest volume at 25°C and 101 k Pa ?
- A 15 g of ethane
 - B 9 g of methane
 - C 12 g of water
 - D 20 g of carbon dioxide
- 20 The reactions show how iodine trichloride is produced



What is the enthalpy change of formation of iodine trichloride?

- A -37 kJ mol^{-1}
 - B -74 kJ mol^{-1}
 - C -81 kJ mol^{-1}
 - D -162 kJ mol^{-1}
- 21 What are the products of the thermal decomposition of calcium nitrate?
- A calcium, oxygen and nitrogen dioxide
 - B calcium oxide, nitrogen dioxide and oxygen
 - C calcium oxide, nitrogen and oxygen
 - D calcium oxide, nitrogen monoxide and oxygen

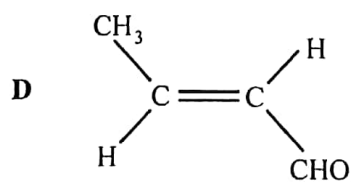
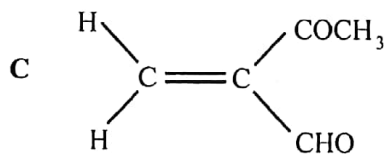
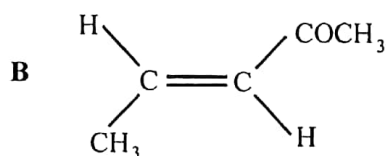
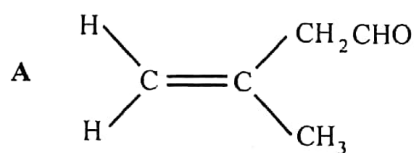
- 22 Structure Y is shown as



How many functional groups are in structure Y?

- A 2
 B 3
 C 4
 D 5
- 23 An organic compound, Y, displays *cis-trans* isomerism and gives a red-brown precipitate with Fehlings solution.

What could be the compound Y?

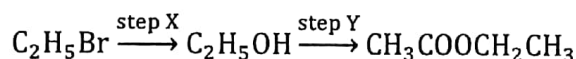


- 24 Which of the given organic compounds are arranged in the correct order of decreasing pK_a values?
- A phenol, ethanol, ethanoic acid, chloroethanoic acid
 B ethanoic acid, chloroethanoic acid, phenol, ethanol
 C chloroethanoic acid, ethanoic acid, phenol, ethanol
 D ethanol, phenol, ethanoic acid, chloroethanoic acid
- 25 The hydrocarbon 2,2 - dimethylbutane undergoes free radical substitution when heated with bromine.

How many free radicals are formed in the propagation step?

- A 4
 B 3
 C 2
 D 1
- 26 Which compound is **not** formed when ethane and chlorine are mixed in the presence of sunlight?
- A $\text{CH}_3\text{CH}_2\text{Cl}$
 B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
 C $\text{CH}_3\text{CHClCHClCH}_3$
 D ClCHCHCl

- 27 Ethylethanoate can be formed from bromoethane in two steps as shown.



Which statement is true about this two step reaction?

- A step X involves electrophilic substitution
 B hot $\text{NaOH}_{(\text{aq})}$ is the reagent for step X
 C step Y is catalysed by dilute $\text{H}_2\text{SO}_{4(\text{aq})}$
 D an acidified aldehyde is the reagent for step Y
- 28 The products obtained by cracking a hydrocarbon are methane, ethane and propene. The mole fraction of ethane in the products is 0.500.

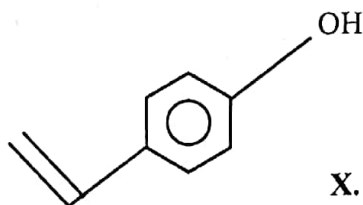
What is the molecular formula of the hydrocarbon?

- A C_6H_{14}
 B C_8H_{18}
 C C_9H_{20}
 D $\text{C}_{11}\text{H}_{24}$

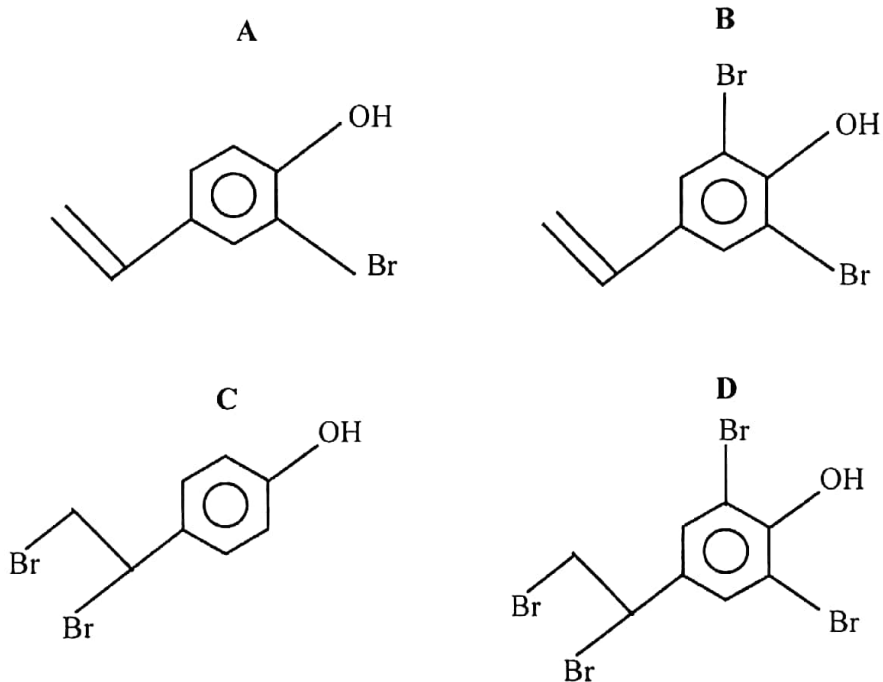
29 Which compound gives fumes of HCl with PCl_5 and NH_3 when heated with $\text{NaOH}_{(\text{aq})}$?

- A $\text{HOCH}_2\text{CONH}_2$
- B $\text{HOCH}_2\text{CH}_2\text{NH}_2$
- C $\text{NH}_2\text{CH}_2\text{CO}_2\text{H}$
- D $\text{HOCH}_2\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$

30 The structure of an organic compound **X** is shown.



What is the product of the reaction between **X** and excess $\text{Br}_{2(\text{aq})}$ at room temperature?



SECTION B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statement(s) is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31 The parameters of an ideal gas equation are given as $pV = nRT$.

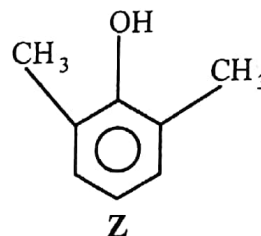
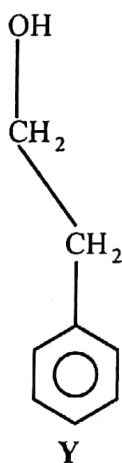
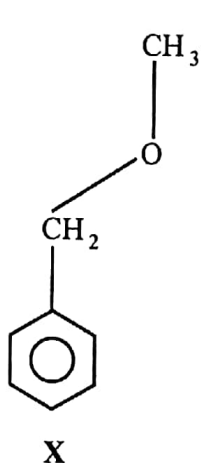
What can be deduced from the gas equation?

1. Pressure is directly proportional to volume at constant temperature.
2. The volume is directly proportional to temperature at constant pressure.
3. The pressure is directly proportional to temperature at constant volume.

- 32 Which statement(s) regarding electronic orbitals of principal quantum numbers 1, 2 and 3 is/are correct?

1. A p-orbital has higher energy than an S orbital on the same principal quantum number.
2. Each P-orbital can contain a maximum of six electrons
3. The three 3p orbitals have slightly different energy levels

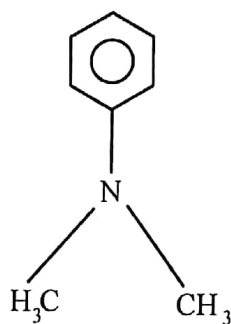
33 Three organic compounds X, Y and Z are shown.



Which statement(s) is/are true about the compounds X, Y and Z.

1. they are isomers
2. they all react with sodium metal
3. they all react with acidified KMnO_4 to produce benzoic acid

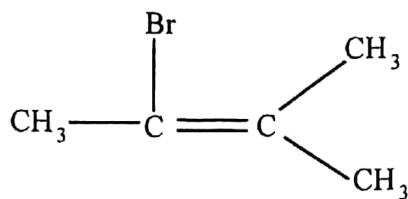
34 The structure of an organic compound is shown.



The compound shown

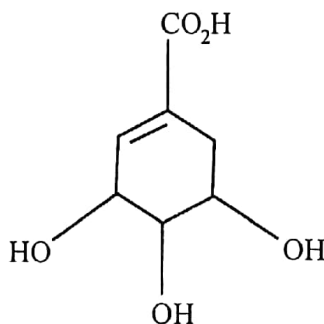
1. does not react with benzene diazonium chloride.
2. decolourises bromine water.
3. does not react with potassium manganate (VII).

- 35 The structure of an organic compound is shown.



Which statement(s) is /are true about the compound?

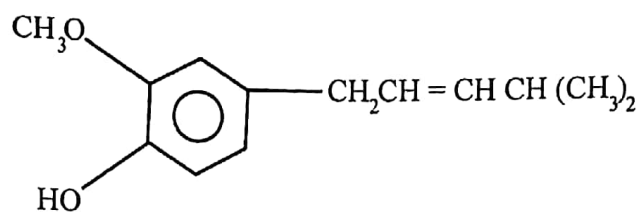
1. the empirical formula is C_5H_9Br
 2. shows *cis-trans* isomerism
 3. show optical isomerism
- 36 An organic compound of the formula $CH_3CHOHCHO$ forms
1. a yellow precipitate with alkaline iodine.
 2. fumes of hydrogen chloride gas with PCl_5 .
 3. an orange precipitate with a solution of 2,4 DNPH.
- 37 The structure of Shikimic acid is shown:



Which statement(s) is/are correct about Shikimic acid?

1. has 5 functional groups
2. reacts with $Br_{2(aq)}$
3. reacts with ethanoyl chloride

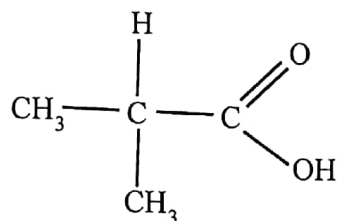
- 38 The organic compound shown was heated under reflux with acidified concentrated KMnO_4 .



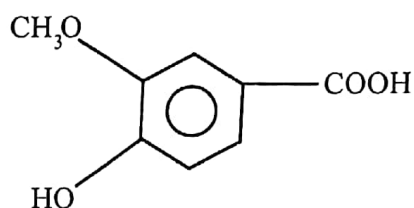
The product(s) of the reaction was/were

1. CO_2

2.



3.



- 39 Which compound(s) show(s) optical isomerism?

1. $\text{CH}_3\text{CHNH}_2\text{CO}_2\text{H}$
2. CH_3CHOHCN
3. $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{C}_6\text{H}_5$

- 40 What is /are the organic product(s) formed when butan-2-ol is heated with dilute acidified potassium manganate (VII).

1. butanone
2. butanoic acid
3. butanal